

SQL Database Management Project

Maham Haider

Database Management (MIS 370) – DePaul University

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SQL Code

```
DROP TABLE CLIENT;  
CREATE TABLE CLIENT(  
    CLIENT_ID INT NOT NULL,  
    CLIENT_FNAME VARCHAR(50),  
    CLIENT_LNAME VARCHAR(50),  
    CLIENT_MINIT VARCHAR(50),  
    REGION_ID VARCHAR(50),  
    ADDRESS VARCHAR(50),  
    CITY VARCHAR(50),  
    STATE VARCHAR(50),  
    ZIP_CODE VARCHAR(50),  
    CLIENT_SSN CHAR(9),  
    EMAIL VARCHAR(50),  
    PHONE VARCHAR(10),  
    PRIMARY KEY (CLIENT_ID)  
);
```

```
DROP TABLE CONTRACT;  
CREATE TABLE CONTRACT(  
    CONTRACT_ID INT  
    CLIENT_ID INT,  
    CONTRACT_START DATE,  
    CONTRACT_END DATE,  
    CONTRACT_AMOUNT INT,  
    PRIMARY KEY (CONTRACT_ID)  
);
```

```
DROP TABLE REGION;  
CREATE TABLE REGION (  
    REGION_ID VARCHAR(50),  
    REGION_NAME VARCHAR(50),  
    PRIMARY KEY (REGION_ID)  
);
```

```
DROP TABLE CONSULTANTS;  
CREATE TABLE CONSULTANTS(  
    CONSULTANT_ID INT,  
    CONSULTANT_FNAME VARCHAR(50),  
    CONSULTANT_LNAME VARCHAR(50),  
    CONSULTANT_MINNIT VARCHAR(50),  
    REGION_ID VARCHAR(50),
```

```

CONSULTANT_EXPERIENCE INT,
CONSULTANT_SSN CHAR(9),
PHONE VARCHAR (10),
EMAIL VARCHAR(50),
PRIMARY KEY (CONSULTANT_ID)
);

DROP TABLE CLASS;
CREATE TABLE CLASS (
    CLASS_ID INT,
    CLASS_NAME VARCHAR(50),
    PRIMARY KEY (CLASS_ID)
);

DROP TABLE CONTRACT_CONSULTANT;
CREATE TABLE CONTRACT_CONSULTANT(
    CONTRACT_ID INT,
    CONSULTANT_ID INT,
    PRIMARY KEY (CONTRACT_ID, CONSULTANT_ID)
);

DROP TABLE CONTRACT_CLASS;
CREATE TABLE CONTRACT_CLASS (
    CONTRACT_ID INT,
    CLASS_ID INT,
    PRIMARY KEY (CONTRACT_ID, CLASS_ID)
);

DROP TABLE CONSULTANT_CLASS;
CREATE TABLE CONSULTANT_CLASS (
    CONSULTANT_ID INT,
    CLASS_ID INT,
    PRIMARY KEY (CONSULTANT_ID, CLASS_ID)
);

Insert into Class (Class_ID, Class_Name)
values          (5, 'Data Analyst');

Insert into Class (Class_ID, Class_Name)
values          (8, 'Business Analyst');

Insert into Class (Class_ID, Class_Name)
values          (4, 'Financial Analyst');

Insert into Class (Class_ID, Class_Name)
values          (3, 'Project Manager');

```

```
Insert into Class (Class_ID, Class_Name)
values          (7, 'Information Technology');
```

```
Insert into Class (Class_ID, Class_Name)
values          (2, 'Digital Marketing');
```

```
Insert into Class (Class_ID, Class_Name)
values          (11, 'Database Design');
```

```
Insert into Class (Class_ID, Class_Name)
values          (9, 'Web Applications');
```

```
Insert into Class (Class_ID, Class_Name)
values          (12, 'Internet Services');
```

```
Insert into Class (Class_ID, Class_Name)
values          (6, 'Database Administration');
```

```
Insert into client (Client_ID, Client_Fname, Client_Minit, Client_Lname, Region_ID, Address,
City, State, Zip_Code, Client_SSN, Email, Phone)
Values          (1, 'Sam', 'M', 'Jong', 'SW', '123 Pine Dr', 'Chicago', 'IL', '60601', '23454029',
'samj@gmail.com', '8384901233');
```

```
Insert into client (Client_ID, Client_Fname, Client_Minit, Client_Lname, Region_ID, Address,
City, State, Zip_Code, Client_SSN, Email, Phone)
Values          (2, 'Ariana', 'N', 'Elahi', 'NW', '432 Midway Dr', 'Brooklyn', 'NY', '90320',
'90293049', 'arianae@gmail.com', '7340592049');
```

```
Insert into client (Client_ID, Client_Fname, Client_Minit, Client_Lname, Region_ID, Address,
City, State, Zip_Code, Client_SSN, Email, Phone)
Values          (3, 'Rihanna', 'R', 'Wade', 'MW', '221 Coast Dr', 'Los Angeles', 'CA', '10234',
'40239048', 'rihannaw@gmail.com', '4103292930');
```

```
Insert into client (Client_ID, Client_Fname, Client_Minit, Client_Lname, Region_ID, Address,
City, State, Zip_Code, Client_SSN, Email, Phone)
Values          (4, 'Huda', 'S', 'Kattan', 'SE', '542 Hemming Dr', 'Austin', 'TX', '43401', '50649302',
'hudak@gmail.com', '1029305678');
```

```
Insert into client (Client_ID, Client_Fname, Client_Minit, Client_Lname, Region_ID, Address,
City, State, Zip_Code, Client_SSN, Email, Phone)
Values          (5, 'Drake', 'T', 'Scott', 'NE', '593 Lake Dr', 'Miami', 'FL', '50492', '60539500',
'drakes@gmail.com', '2039404592');
```

Insert into client (Client_ID, Client_Fname, Client_Minit, Client_Lname, Region_ID, Address, City, State, Zip_Code, Client_SSN, Email, Phone)

Values (6, 'Amr', 'B', 'Diab', 'NE', '500 Maine Dr', 'Portland', 'ME', '20363', '29304950', 'amrd@gmail.com', '3044959203');

Insert into client (Client_ID, Client_Fname, Client_Minit, Client_Lname, Region_ID, Address, City, State, Zip_Code, Client_SSN, Email, Phone)

Values (7, 'Wayne', 'P', 'Tristian', 'SW', '234 Sunset Dr', 'Dearborn', 'MI', '30485', '96540699', 'waynet@gmail.com', '5020394023');

Insert into client (Client_ID, Client_Fname, Client_Minit, Client_Lname, Region_ID, Address, City, State, Zip_Code, Client_SSN, Email, Phone)

Values (8, 'Travis', 'A', 'Rocky', 'MW', '902 River Dr', 'Denver', 'CO', '90302', '10293029', 'travisr@gmail.com', '6039203945');

Insert into client (Client_ID, Client_Fname, Client_Minit, Client_Lname, Region_ID, Address, City, State, Zip_Code, Client_SSN, Email, Phone)

Values (9, 'Kylie', 'K', 'Jenner', 'SE', '433 County Dr', 'Phoenix', 'AZ', '83458', '80203940', 'kyliej@gmail.com', '7063040495');

Insert into client (Client_ID, Client_Fname, Client_Minit, Client_Lname, Region_ID, Address, City, State, Zip_Code, Client_SSN, Email, Phone)

Values (10, 'Kourtney', 'Q', 'Thompson', 'MW', '975 Salt Dr', 'Las Vegas', 'NV', '70425', '50490346', 'kourtneyt@gmail.com', '9034950395');

Insert into Consultants (Consultant_ID, Consultant_Fname, Consultant_Lname, Consultant_Minit, Region_ID, Consultant_Experience, Consultant_SSN, Phone, Email)

values (234, 'Ariana', 'Elahi', 'N', 'NW', 5, '90293049', '7340592049', 'arianae@gmail.com')

Insert into Consultants (Consultant_ID, Consultant_Fname, Consultant_Lname, Consultant_Minit, Region_ID, Consultant_Experience, Consultant_SSN, Phone, Email)

values (123, 'Huda', 'Kattan', 'S', 'SE', 10, '50649048', '1029305678', 'hudak@gmail.com')

Insert into Consultants (Consultant_ID, Consultant_Fname, Consultant_Lname, Consultant_Minit, Region_ID, Consultant_Experience, Consultant_SSN, Phone, Email)

values (456, 'Travis', 'Rocky', 'A', 'MW', 3, '10293029', '6039203945', 'travisr@gmail.com')

Insert into Consultants (Consultant_ID, Consultant_Fname, Consultant_Lname, Consultant_Minit, Region_ID, Consultant_Experience, Consultant_SSN, Phone, Email)

values (923, 'Kylie', 'Jenner', 'K', 'SE', 6, '80203940', '7063040495', 'kyliej@gmail.com')

Insert into Consultants (Consultant_ID, Consultant_Fname, Consultant_Lname,
Consultant_Minnit, Region_ID, Consultant_Experience, Consultant_SSN, Phone, Email)

values (540, 'Drake', 'Scott', 'T', 'NE', 8, '60539500', '2039404592',
'drakes@gmail.com')

Insert into Consultants (Consultant_ID, Consultant_Fname, Consultant_Lname,
Consultant_Minnit, Region_ID, Consultant_Experience, Consultant_SSN, Phone, Email)

values (930, 'Sam', 'Jong', 'M', 'SW', 2, '23454029', '9394901233', 'samj@gmail.com')

Insert into Consultants (Consultant_ID, Consultant_Fname, Consultant_Lname,
Consultant_Minnit, Region_ID, Consultant_Experience, Consultant_SSN, Phone, Email)

values (650, 'Rihanna', 'Wade', 'R', 'MW', 13, '40239048', '4103292930',
'rihannaw@gmail.com')

Insert into Consultants (Consultant_ID, Consultant_Fname, Consultant_Lname,
Consultant_Minnit, Region_ID, Consultant_Experience, Consultant_SSN, Phone, Email)

values (340, 'Wayne', 'Tristian', 'P', 'SW', 4, '96540699', '5020394023',
'waynet@gmail.com')

Insert into Consultants (Consultant_ID, Consultant_Fname, Consultant_Lname,
Consultant_Minnit, Region_ID, Consultant_Experience, Consultant_SSN, Phone, Email)

values (209, 'Kourtney', 'Thompson', 'Q', 'MW', 9, '50490346', '9034950395',
'kourtneyt@gmail.com')

Insert into Consultants (Consultant_ID, Consultant_Fname, Consultant_Lname,
Consultant_Minnit, Region_ID, Consultant_Experience, Consultant_SSN, Phone, Email)

values (756, 'Amr', 'Diab', 'B', 'NE', 5, '29304950', '3044959203', 'amrd@gmail.com')

Insert into Region (Region_ID, Region_Name)

values ('SW', 'Southwest');

Insert into Region (Region_ID, Region_Name)

values ('NW', 'Northwest');

Insert into Region (Region_ID, Region_Name)

values ('MW', 'Midwest');

Insert into Region (Region_ID, Region_Name)

values ('SE', 'Southeast');

Insert into Region (Region_ID, Region_Name)

values ('NE', 'Northeast');

Insert into Contract (Contract_ID, Client_ID, Contract_Start, Contract_End, Contract_Amount)

values (23, 1, '2023-01-01', '2026-01-15', 3500649);

Insert into Contract (Contract_ID, Client_ID, Contract_Start, Contract_End, Contract_Amount)

values (27, 2, '2022-01-15', '2025-03-03', 9034056);

Insert into Contract (Contract_ID, Client_ID, Contract_Start, Contract_End, Contract_Amount)

values (35, 3, '2022-04-15', '2025-05-08', 9008601);

Insert into Contract (Contract_ID, Client_ID, Contract_Start, Contract_End, Contract_Amount)

values (15, 4, '2022-02-14', '2026-02-16', 2001305);

Insert into Contract (Contract_ID, Client_ID, Contract_Start, Contract_End, Contract_Amount)

values (40, 5, '2022-06-23', '2026-06-23', 6213234);

Insert into Contract (Contract_ID, Client_ID, Contract_Start, Contract_End, Contract_Amount)


```
values (42, 6, '2024-07-23', '2027-12-05', 5345059);
```

```
Insert into Contract (Contract_ID, Client_ID, Contract_Start, Contract_End, Contract_Amount)
```

```
values (29, 7, '2021-08-29', '2024-05-07', 8039945);
```

```
Insert into Contract (Contract_ID, Client_ID, Contract_Start, Contract_End, Contract_Amount)
```

```
values (37, 8, '2023-11-23', '2026-11-24', 8909234);
```

```
Insert into Contract (Contract_ID, Client_ID, Contract_Start, Contract_End, Contract_Amount)
```

```
values (60, 9, '2022-10-31', '2025-12-25', 7345678);
```

```
Insert into Contract (Contract_ID, Client_ID, Contract_Start, Contract_End, Contract_Amount)
```

```
values (53, 10, '2024-01-12', '2027-11-12', 9998221);
```

```
Insert into Consultant_Class (Consultant_ID, Class_ID)
```

```
values (234, 5);
```

```
Insert into Consultant_Class (Consultant_ID, Class_ID)
```

```
values (123, 8);
```

```
Insert into Consultant_Class (Consultant_ID, Class_ID)
```

```
values (456, 4);
```

```
Insert into Consultant_Class (Consultant_ID, Class_ID)
```

```
values (923, 3);
```

```
Insert into Consultant_Class (Consultant_ID, Class_ID)
values (540, 7);
```

```
Insert into Consultant_Class (Consultant_ID, Class_ID)
values (930, 2);
```

```
Insert into Consultant_Class (Consultant_ID, Class_ID)
values (650, 11);
```

```
Insert into Consultant_Class (Consultant_ID, Class_ID)
values (340, 9);
```

```
Insert into Consultant_Class (Consultant_ID, Class_ID)
values (209, 12);
```

```
Insert into Consultant_Class (Consultant_ID, Class_ID)
values (756, 6);
```

```
Insert into Contract_Class (Contract_ID, Class_ID)
values (23, 5);
```

```
Insert into Contract_Class (Contract_ID, Class_ID)
values (27, 8);
```

```
Insert into Contract_Class (Contract_ID, Class_ID)
values (35, 4);
```

```
Insert into Contract_Class (Contract_ID, Class_ID)
values (15, 3);
```

```
Insert into Contract_Class (Contract_ID, Class_ID)
values (40, 7);
```

```
Insert into Contract_Class (Contract_ID, Class_ID)
values (42, 2);
```

```
Insert into Contract_Class (Contract_ID, Class_ID)
values (29, 11);
```

```
Insert into Contract_Class (Contract_ID, Class_ID)
values (37, 9)
```

```
Insert into Contract_Class (Contract_ID, Class_ID)
values (60, 12)
```

```
Insert into Contract_Class (Contract_ID, Class_ID)
values (53, 6)
```

```
Insert into Contract_Constant (Contract_ID, Consultant_ID)
values (23, 234);
```

```
Insert into Contract_Constant (Contract_ID, Consultant_ID)
values (27, 123);
```

```
Insert into Contract_Constant (Contract_ID, Consultant_ID)
values (35, 456);
```

```
Insert into Contract_Constant (Contract_ID, Consultant_ID)
values (15, 923);
```

```
Insert into Contract_Constant (Contract_ID, Consultant_ID)
values (40, 540);
```

```
Insert into Contract_Constant (Contract_ID, Consultant_ID)
values (42, 930);
```

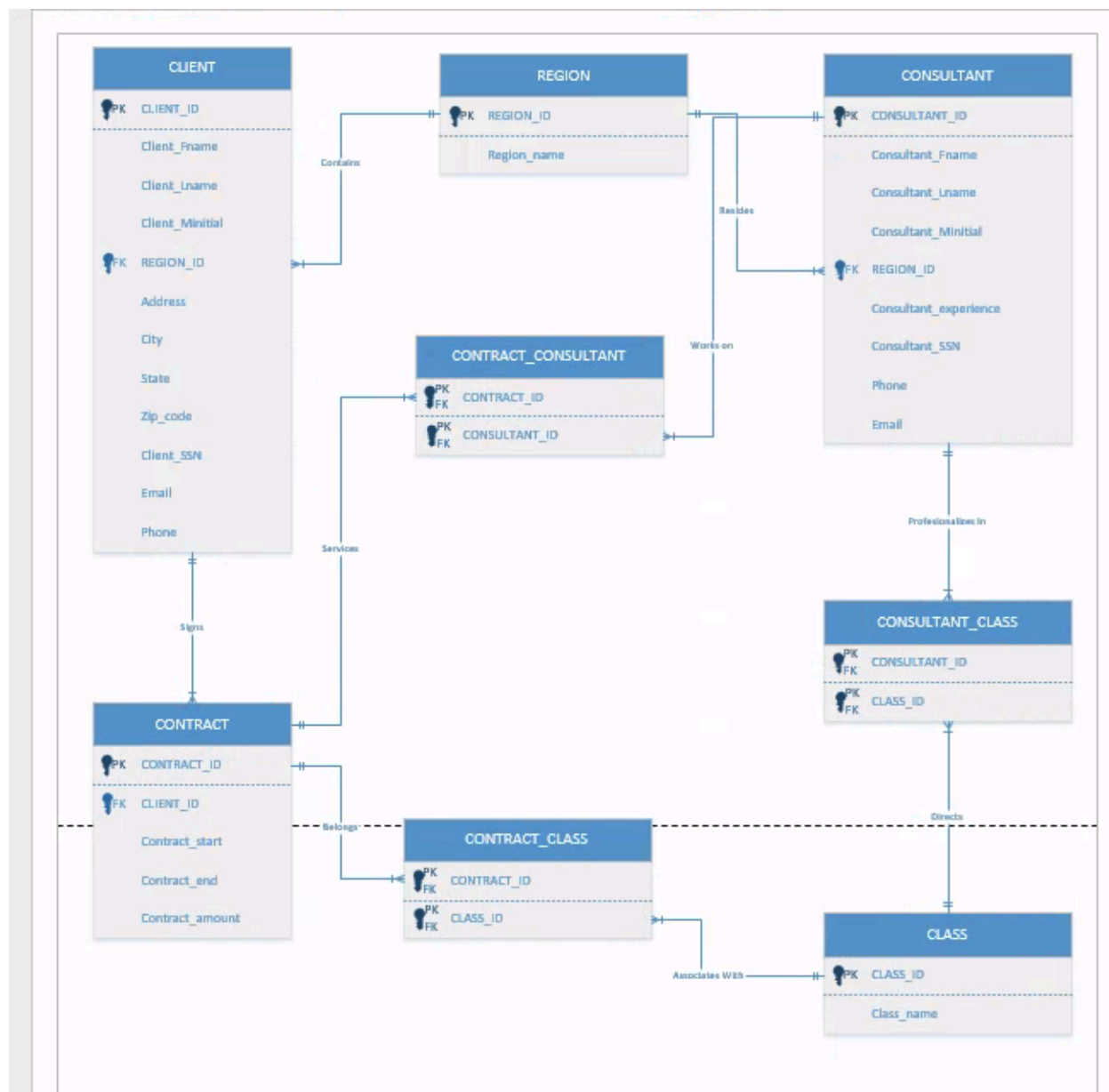
```
Insert into Contract_Constant (Contract_ID, Consultant_ID)
values (29, 650);
```

```
Insert into Contract_Constant (Contract_ID, Consultant_ID)
values (37, 340);
```

```
Insert into Contract_Constant (Contract_ID, Constant_ID)
values                          (60, 209);
```

```
Insert into Contract_Constant (Contract_ID, Constant_ID)
values                          (53, 756);
```

Entity Relationship Diagram



3NF Relational Schema

CLIENT(CLIENT_ID, CLIENT_FNAME, CLIENT_LNAME,
CLIENT_MINICIAL, REGION_ID, ADDRESS, CITY, STATE,
ZIP_CODE, CLIENT_SSN, EMAIL, PHONE)

CONTRACT(CONTRACT_ID, CLIENT_ID, CONTRACT_START,
CONTRACT_END, CONTRACT_AMOUNT)

REGION(REGION_ID, REGION_NAME)

CONSULTANT(CONSULTANT_ID, CONSULTANT_FNAME,
CONSULTANT_LNAME, CONSULTANT_MINICIAL, REGION_ID,
CONSULTANT_EXPERIENCE, CONSULTANT_SSN, PHONE,
EMAIL)

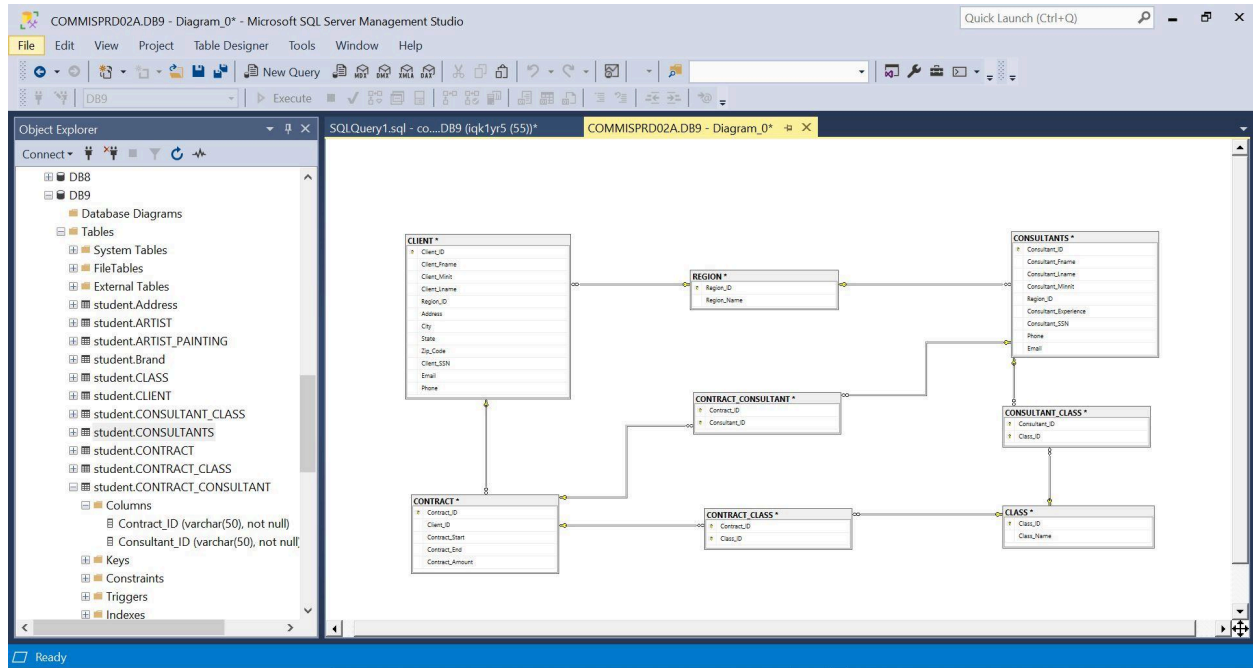
CLASS(CLASS_ID, CLASS_NAME)

CONTRACT_CONSULTANT(CONTRACT_ID, CONSULTANT_ID)

CONTRACT_CLASS(CONTRACT_ID, CLASS_ID)
.....

CONSULTANT_CLASS(CONSULTANT_ID, CLASS_ID)
.....

Rational Diagram

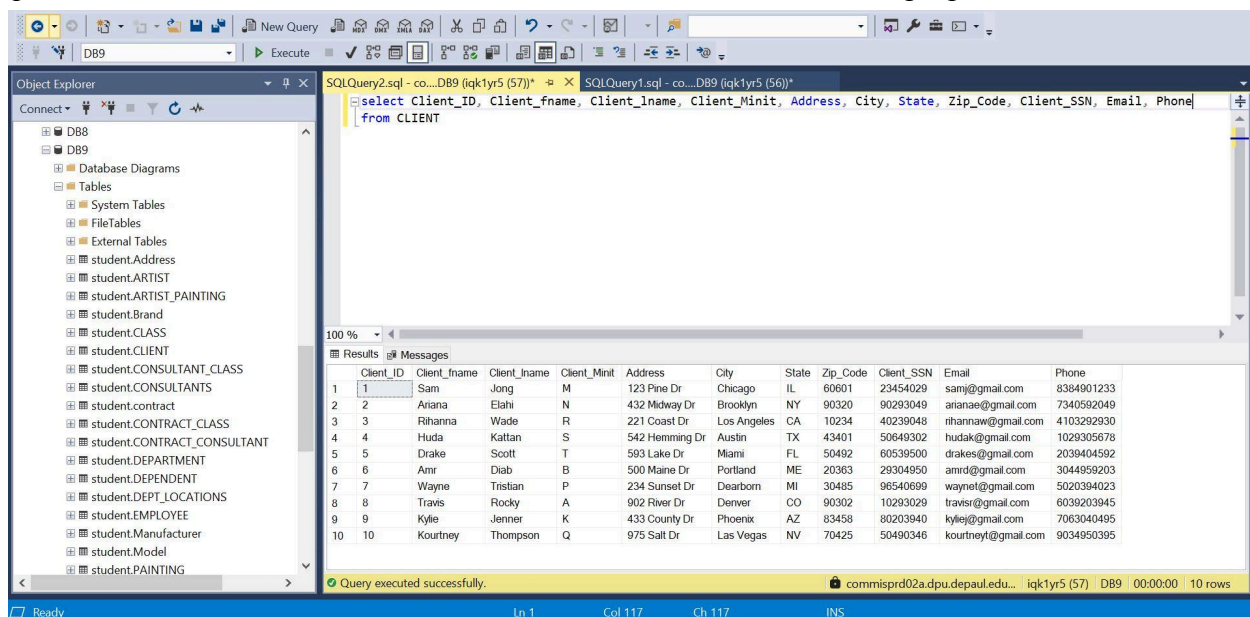


Reports

1- This report provides a comprehensive overview of each consultant's personal and contact information. By doing so, we help the managers easily identify who the consultants are and how to reach them. This insight is extremely valuable for efficient communication when connecting the consultants with the clients.

The screenshot shows a SQL Server Enterprise Manager interface. The Object Explorer on the left displays a database named 'DB9' with various tables. The main window shows a query executed successfully, returning 10 rows of consultant data. The status bar at the bottom indicates 'Query executed successfully.' and 'commisprd02a.dpu.depaul.edu... | iqk1yr5 (56) DB9 00:00:00 10 rows'.

2- The record of clients is extremely important for a manager. It includes essential client information, including their names, Social Security Numbers, email addresses, and their phone numbers. This report offers a centralized view of client contact and personal details, allowing quick access to client information for communication and verification purposes.



The screenshot shows a SQL Server Enterprise Manager interface. The Object Explorer on the left displays a database named 'DB9' with various tables. The main window shows a query executed successfully, returning 10 rows of client data. The status bar at the bottom indicates 'Query executed successfully.' and 'commisprd02a.dpu.depaul.edu... | iqk1yr5 (57) DB9 00:00:00 10 rows'.

Client_ID	Client_fname	Client_lname	Client_Mininit	Address	City	State	Zip_Code	Client_SSN	Email	Phone
1	Sam	Jong	M	123 Pine Dr	Chicago	IL	60601	23454029	samj@gmail.com	8384901233
2	Ariana	Elahi	N	432 Midway Dr	Brooklyn	NY	90320	90293049	arianae@gmail.com	7340592049
3	Rihanna	Wade	R	221 Coast Dr	Los Angeles	CA	10234	40239048	rihannaw@gmail.com	4103292930
4	Huda	Kattan	S	542 Hemming Dr	Austin	TX	43401	506490302	hudak@gmail.com	1029305678
5	Drake	Scott	T	593 Lake Dr	Miami	FL	50492	60539500	drakes@gmail.com	2039404592
6	Amr	Diab	B	500 Maine Dr	Portland	ME	20363	29304950	amrd@gmail.com	3044959203
7	Wayne	Tristian	P	234 Sunset Dr	Dearborn	MI	30485	96540699	waynet@gmail.com	5020394023
8	Travis	Rocky	A	902 River Dr	Denver	CO	90302	10293029	travisr@gmail.com	6039203945
9	Kyle	Jenner	K	433 County Dr	Phoenix	AZ	83458	80203940	kylej@gmail.com	7063040495
10	Kourtney	Thompson	Q	975 Salt Dr	Las Vegas	NV	70425	50490346	kourtneyt@gmail.com	9034950395

3- This report provides an overview of all the contracts, including the ID, start and end dates, the contract amount, and which clients are involved. By reviewing this information, managers can get betting insight of business performance trends. They can identify the strong and underperforming areas, allowing strategic adjustments to enhance business growth.

The screenshot shows the SQL Server Enterprise Manager interface. The Object Explorer on the left displays the database structure for 'DB9'. The central pane shows a SQL query for 'SQLQuery3.sql' that joins the 'CONTRACT' table with the 'CLIENT' table. The 'Results' pane at the bottom displays the query output as a table with 10 rows and 7 columns.

	contract_ID	Contract_Start	Contract_End	Contract_amount	client_fname	client_lname	client_minit
1	23	2023-01-01	2026-01-15	3500649	Sam	Jong	M
2	27	2022-01-15	2025-03-03	9034056	Ariana	Elahi	N
3	35	2022-04-15	2025-05-08	9008601	Rihanna	Wade	R
4	15	2022-02-14	2026-02-16	2001305	Huda	Kattan	S
5	40	2022-06-23	2026-06-23	6213234	Drake	Scott	T
6	42	2024-07-23	2027-12-05	5345059	Amr	Diab	B
7	29	2021-08-29	2024-05-07	8039945	Wayne	Tristian	P
8	37	2023-11-23	2026-11-24	8909234	Travis	Rocky	A
9	60	2022-10-31	2025-12-25	7345678	Kylie	Jenner	K
10	53	2024-01-12	2027-11-12	9998221	Kourtney	Thompson	Q

4- This report provides a summary of the number of clients and consultants within each region. By comparing client demand to consultant availability across regions, managers can identify areas where resources may be under-utilized or over-utilized. Doing this enables strategic adjustments to improve service coverage and client satisfaction.

The screenshot shows the SQL Server Enterprise Manager interface. The Object Explorer on the left displays the database structure for 'DB9'. The central pane shows a SQL query for 'SQLQuery4b.sql' that counts the number of clients and consultants for each region. The 'Results' pane at the bottom displays the query output as a table with 5 rows and 4 columns.

	client_region	number_of_clients	consultant_region	number_of_consultant
1	Midwest	9	Midwest	9
2	Northeast	4	Northeast	1
3	Northwest	1	Northwest	4
4	Southeast	4	Southeast	4
5	Southwest	4	Southwest	4

5- This report provides an overview of consultants by region, including their ID, names, and what class they teach. Doing this allows managers to see which consultants are available in each region, and their specialties. Organizing consultants by region and expertise supports effective resource allocation and enhances client-consulting matching by aligning regional demand with available skill sets.

The screenshot shows the SQL Server Enterprise Manager interface. The Object Explorer on the left displays the database structure for 'DB9'. The central pane shows a SQL query executed successfully, displaying the results of a query that joins the REGION, CONSULTANTS, CONSULTANT_CLASS, and CLASS tables. The results are shown in a table with 10 rows and 5 columns: Region_Name, Consultant_ID, consultant_fname, consultant_lname, and Class_Name.

Region_Name	Consultant_ID	consultant_fname	consultant_lname	Class_Name
Northwest	234	Ariana	Elahi	Data Analyst
Southeast	123	Huda	Kattan	Business Analyst
Midwest	456	Travis	Rocky	Financial Analyst
Southeast	923	Kylie	Jenner	Project Manager
Northeast	540	Drake	Scott	Information Technology
Southwest	930	Sam	Jong	Digital Marketing
Midwest	650	Rihanna	Wade	Database Design
Southwest	340	Wayne	Tristian	Web Applications
Midwest	209	Kourtney	Thompson	Internet Services
Northeast	756	Amr	Diab	Database Administration

6- This report presents contract details along with which clients were included in the contract. The report displays which contracts are still active and which have been completed. This provides managers with a clear view of current client engagements and completed projects, supporting informed decision-making and client satisfaction / relationship management.

The screenshot shows the SQL Server Enterprise Manager interface. The Object Explorer on the left displays the database structure for 'DB9'. The central pane shows a SQL query executed successfully, displaying the results of a query that joins the CONTRACT and CLIENT tables. The results are shown in a table with 10 rows and 7 columns: contract_id, contract_start, contract_end, contract_amount, client_fname, client_lname, and Contract_Status.

contract_id	contract_start	contract_end	contract_amount	client_fname	client_lname	Contract_Status
23	2023-01-01	2026-01-15	3500649	Sam	Jong	Active Contract
27	2022-01-15	2025-03-03	9034056	Ariana	Elahi	Active Contract
35	2022-04-15	2025-05-08	9008601	Rihanna	Wade	Active Contract
4	2022-02-14	2026-02-16	2001305	Huda	Kattan	Active Contract
15	2022-06-23	2026-06-23	6213234	Drake	Scott	Active Contract
42	2024-07-23	2027-12-05	5345059	Amr	Diab	Active Contract
29	2021-08-29	2024-05-07	8038945	Wayne	Tristian	Expired Contract
37	2023-11-23	2026-11-24	8908234	Travis	Rocky	Active Contract
60	2022-10-31	2025-12-25	7345678	Kylie	Jenner	Active Contract
53	2024-01-12	2027-11-12	9998221	Kourtney	Thompson	Active Contract

7- This report displays the total contract revenue generated in each region. It gives managers insights into regional performance, helping to identify high-revenue regions, and which regions could use improvement. Managers could use this information to assess growth-opportunities across all the locations and drastically improve business.

The screenshot shows the SQL Server Enterprise Manager interface. The Object Explorer on the left displays the database structure for 'DB9'. The central pane shows a SQL query for 'SQLQuery7.sql' that calculates the total contract amount for each region. The query is as follows:

```
select r.region_name, sum(c.contract_amount) as total_contract_amount
from CONTRACT c
join client cl on (c.client_id = cl.client_id)
join region r on (cl.region_id = r.region_id)
group by r.region_name;
```

The Results pane shows the output of the query, displaying the region name and the total contract amount for five regions:

region_name	total_contract_amount
Midwest	27016056
Northeast	11558293
Northwest	9034056
Southeast	9348983
Southwest	11540594

The status bar at the bottom indicates 'Query executed successfully.' and '5 rows'.

8- This report provides the average experience level of all consultants, with the average combined experience amounting to six years. It offers managers a summary of the overall expertise level within the consulting team, helping to assess the team's capacity to handle complex projects. It also helps the company prove credibility when searching for new clients. This number can assist managers in making more informed decisions about project assignments.

The screenshot shows the SQL Server Enterprise Manager interface. The Object Explorer on the left displays the database structure for 'DB9'. The central pane shows a SQL query for 'SQLQuery8.sql' that calculates the average experience level of all consultants. The query is as follows:

```
select
avg(cast(consultant_experience as INT)) as average_experience
from CONSULTANTS;
```

The Results pane shows the output of the query, displaying the average experience level:

average_experience
6

The status bar at the bottom indicates 'Query executed successfully.' and '1 rows'.

9- This report provides a breakdown of the number of contracts for each class. Since each contract is tied to a single service class, this helps managers track the distribution of business across different service areas, identify trends, and make informed decisions about future resource allocations.

The screenshot shows the SQL Server Enterprise Manager interface. The Object Explorer on the left displays the database structure for DB9, including tables like student.CONTRACT_CLASS and student.CONTRACT. The central query window shows the following SQL query:

```
select c1.class_name, count(cc.contract_id) as contract_count
from CONTRACT_CLASS cc
join class c1 on (cc.class_id = c1.class_id)
group by c1.class_name;
```

The Results pane at the bottom displays the following data:

class_name	contract_count
1 Business Analyst	1
2 Data Analyst	1
3 Database Administration	1
4 Database Design	1
5 Digital Marketing	1
6 Financial Analyst	1
7 Information Technology	1
8 Internet Services	1
9 Project Manager	1
10 Web Applications	1

The status bar at the bottom indicates "Query executed successfully." and "10 rows".

10- This report lists each consultant, their ID, and the class they are assigned to, along with the corresponding class ID and name. It provides an overview of consultant expertise by class, allowing managers to efficiently match consultants with projects and ensure that clients receive the appropriate expertise for their needs.

The screenshot shows the SQL Server Enterprise Manager interface. The Object Explorer on the left displays the database structure for DB9, including tables like student.CONTRACT_CLASS and student.CONTRACT. The central query window shows the following SQL query:

```
select cc.consultant_id, cc.class_id, c.class_name, consultant_fname, consultant_lname
from CONSULTANT_CLASS cc
join class c on (cc.class_id = c.class_id)
join CONSULTANTS con on (cc.consultant_id = con.consultant_id);
```

The Results pane at the bottom displays the following data:

consultant_id	class_id	class_name	consultant_fname	consultant_lname
1 234	5	Data Analyst	Aniana	Elahi
2 123	8	Business Analyst	Huda	Kaltan
3 456	4	Financial Analyst	Travis	Rocky
4 923	3	Project Manager	Kylie	Jenner
5 540	7	Information Technology	Drake	Scott
6 930	2	Digital Marketing	Sam	Jong
7 650	11	Database Design	Rihanna	Wade
8 340	9	Web Applications	Wayne	Tristian
9 209	12	Internet Services	Kourtney	Thompson
10 756	6	Database Administration	Amr	Diab

The status bar at the bottom indicates "Query executed successfully." and "10 rows".